

SEBASTIAN HORTA

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EDUCATION

Columbia University (SEAS), NY — Computer Science B.S., Minor: Electrical Engineering Aug 2023 - May 2026

- Relevant coursework: Artificial Intelligence, Systems Programming, UI Design, Linear Algebra, Fundamentals of Computer Systems, Computer Science Theory, Data Structures, Probability
- Resident Adviser (Engagement Coordinator) (2024-Present): Organize community-building events

Farmingdale State College (FSC), NY — Computer Science B.S. (Transferred) Aug 2022 - May 2023

- CGPA: 4.0 (President's List), Collegiate Science and Technology Entry Program (CSTEP)
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SKILLS/LANGUAGES/TECHNOLOGIES

- **Programming**: Java, Python, C, JavaScript, TypeScript, HTML/CSS | **Frontend**: React, Next.js, AngularJS, Bootstrap
 - **Backend & APIs**: Flask, Node.js, FastAPI, REST APIs, MySQL, backend architecture
 - **AI/ML & Data**: TensorFlow, scikit-learn, face_recognition, AI pipelines, OpenAI API, Python scripting
 - **DevOps & Platforms**: GCP, Unix/Linux, web servers, CI/CD, Git | **Other**: Godot, customtkinter, PIL
 - **Languages**: English (Fluent), Spanish (Fluent), French (Basic)
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PROJECTS & EXPERIENCE

- **Web Archiver Tool (2025)**: Developed a React + FastAPI app using async crawling, recursive parsing, asset bundling, and RESTful APIs to capture timestamped static site snapshots with dynamic URL rewriting and a responsive UI for archive navigation.
- **Face Sorter Pro (2025)**: Built a Python desktop app with face_recognition, customtkinter, and PIL to detect, timestamp, and filter video frames by faces. Streamlined media review with automated face-based filtering and image processing.
- **Personalized Gait Analysis Using Machine Learning (2024-25)**: Built an end-to-end ML pipeline analyzing VR sensor data to detect gait events (~85% accuracy) via advanced preprocessing, feature engineering, and Random Forest tuning. Integrated real-time AI feedback.
- **Web Server Development (2024)**: Built a secure, multithreaded HTTP/1.0 server in C supporting concurrent GET requests, with DevOps-focused logging, resource management, and protocol compliance.
- **DevFest Hackathon Sustainability Project (2024)**: Led full-stack development of an API-driven food inventory tracker using Python and GCP, reducing spoilage by 10% through real-time alerts via integrated REST APIs and notification systems.

ACTIVITIES & ORGANIZATIONS

- **Society of Hispanic Professional Engineers (SHPE) at Columbia University - Committee Board Member (2023-Present)**: Organized professional and cultural events, including elevator pitch workshops, boosting inclusivity and SHPE visibility.
 - **Collegiate Science and Technology Entry Program (CSTEP) at FSC - Member (2022-23)**: Collaborated on a senior-led engineering project as a freshman, demonstrating early initiative and teamwork.
 - **Orthodox Presbyterian Church - Member/Volunteer (2018-23)**: Participated in local service initiatives.
 - **Other Involvement (2023-Present)**: Columbia IEEE, ColorStack, CU ADI, CU BJJ Club, Sabor Dance Club
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AWARDS

- Anders Scholarship (2024) • Columbia University Nathaniel Arbiter Scholarship (2024) • FSC President's List (2022-23) • NYS Scholarship for Academic Excellence (2022)